Introduction to climatology: comparing variability

Supplement BG09 · introductory draft

Weather and climate

Weather is the atmospheric state at a time and place. Climate describes characteristics of weather over a specified region and period, including distributions, variability, seasonality and extremes as well as averages. A single hot or cold day cannot establish a long-term climate change.

Space and time

Climate varies from day to day, across seasons, from year to year, over decades and longer. A forest, station, region and whole globe represent different spatial scopes. Elevation and slope aspect matter in mountain environments.

Two regions with the same annual mean can have different summer-to-winter contrasts or wet and dry seasons. Even with the same monthly mean, daily temperature extremes or short intense rainfall can differ. Select an aggregation that fits the question.

Normals and reference periods

One WMO climatological standard normal is the 30-year period 1991–2020. The Japan Meteorological Agency uses the corresponding 2020 normal during 2021–2030. Thirty years is a standard reference convention, not the only possible definition of climate.

Other baselines can serve long-term assessments. A change of baseline alters an anomaly calculated from the same observation. Always name the period, location and month.

Anomaly and ratio

Temperature anomaly = observed value − reference value, in °C. Precipitation as a percent of normal = observed precipitation ÷ reference precipitation × 100, in percent. A ratio is undefined when its reference value is zero.

Even with an identical reference period, missing data and changes in station history affect interpretation. A 2020 normal names the final year of the baseline, not an observation from 2020 alone.

Connecting observations and tree rings

Calibrate a model relating ring indices to observations, for example spring temperature. Test it against another period or independent evidence and report location, season, units and sample depth. Check station moves, missing observations, instrument changes and data processing.

Changing responses

At some sites the relationship of tree growth to temperature weakens or trends diverge over time: this is called the divergence problem. Responses differ across species and sites; no single cause can be assumed.

Tree rings are proxies, not direct climate measurements. Reconstructions estimate past conditions using observations and models. They differ from projections of future climate.

Season, region and long-term variation

An increase in annual mean temperature need not be equally large in every season. The season relevant to growth varies by species and site. Examine monthly or seasonal conditions using physiology and prior research, rather than relying only on annual values.

Means and extremes

A monthly temperature mean does not directly reveal the frequency of daily heat or cold extremes. Equal monthly precipitation can result from a short intense storm or many light rains, with potentially different effects on soil and runoff. Distinguish means, distributions and extremes.

Show year-to-year variability separately from longer trends. A fitted trend over a short interval can be sensitive to its start and end years.

Changing the baseline: fictional example

Suppose observed April temperature is 13°C. If baseline A is 11°C, the anomaly is +2°C; if baseline B is 12°C, it is +1°C. The observation is identical: only the reference changes. Check the baseline of each figure before comparison.

Does a station represent a forest?

Even a nearby station may differ in elevation, aspect, distance and surrounding land use. Check station moves, instrument changes, missing data and urban development. Station histories and comparisons with other stations or gridded datasets offer additional evidence.

Limits of reconstruction

Document the periods, seasons and processing used for calibration and testing. Agreement in one period does not guarantee equal skill in another. Keep uncertainty in calendar dating separate from uncertainty in estimated climate.

Exercise: fictional values

At a station April mean temperature was 13°C and the reference mean was 11°C. April precipitation was 120 mm and the normal was 100 mm. Calculate the temperature anomaly and precipitation as a percent of normal.

Answer: +2°C and 120%. Normals depend on the reference period; one month does not establish a long-term climate trend.

Continue learning

See Lesson 7 on dendroclimatology, BG08 on meteorology and BG03 on mathematics and statistics.

WMO: climate normals · JMA: climate normals · Ohyama et al. (2013): tree rings and spring temperature · Wilson et al. (2007): temperature divergence